



BANGLADESH UNIVERSITY OF BUSINESS AND TECHNOLOGY

Department of Computer Science and Engineering

CSE 400: Software Development IV

PROJECT PROPOSAL

CropIQ: AI-Powered Agriculture Intelligence Platform

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Submitted By

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1. Problem Statement

Bangladesh is an agriculture-dependent nation where over 40% of the workforce is engaged in farming, yet farmers and agribusiness operators continue to face critical challenges: lack of access to real-time crop disease diagnosis, inability to track live market prices, and absence of personalized farming advisory tools in the Bengali language. Several existing digital agriculture platforms in Bangladesh attempt to address fragments of these problems individually, but none provides a unified, complete, and freely accessible solution for Bangladeshi farmers in the Bengali language.

The specific problems this project addresses are:

- **Crop Disease Identification:** Farmers cannot easily identify crop diseases without visiting an agricultural extension officer, which is time-consuming and often inaccessible in rural areas. Delayed diagnosis leads to widespread crop losses.
- **Market Price Unawareness:** Farmers often sell produce below fair market value because they lack access to real-time commodity prices from markets across Bangladesh.
- **No Personalized Advisory:** Weather-dependent planting, harvesting, and irrigation decisions are made without data-driven guidance tailored to the farmer's specific location and crop type.
- **Language Barrier:** Most existing agricultural AI tools are in English only, making them inaccessible to rural Bangladeshi farmers who primarily communicate in Bengali.
- **Fragmented Tools:** Existing Bangladeshi digital agriculture platforms each address one isolated problem, some offer AI-based crop disease advice, others provide weather-based crop recommendations, and some provide plant disease image analysis, but none integrates all capabilities into a single unified platform. CropIQ is built from scratch to close this gap with a superior, unified architecture.

2. Objectives

To design and develop a unified, bilingual, AI-powered web platform providing Bangladeshi farmers with free, real-time tools for crop disease diagnosis, market price tracking, weather-based advisory, and Bengali-language farming Q&A, all in a single application.

Specific Objectives

- To develop an AI Crop Disease Detector that allows users to upload a photo of an affected crop and receive a disease diagnosis with treatment recommendations in Bengali and English using image recognition AI.
- To build a Live Market Price Tracker that displays current commodity prices (rice, wheat, vegetables, etc.) across major Bangladesh markets, updated daily via automated data collection.
- To implement a Weather-Based Crop Advisory system that fetches 7-day weather forecasts for the user's location and generates AI-powered planting and harvesting recommendations.

- To create an AI Agriculture Chatbot that answers farming questions in Bengali using a curated knowledge base of Bangladesh agriculture information.
- To build a User Account System allowing farmers to save tool results, receive personalized alerts, and track their query history.
- To ensure the application is mobile-responsive and accessible on low-bandwidth connections, targeting Android smartphone users in rural Bangladesh.

3. Existing System Analysis

A market analysis of existing Bangladeshi agri-tech platforms reveals strong demand for AI agriculture tools, yet no single platform addresses all farmer needs:

Platform		What It Offers	Key Gap
AGrocery (agrocerybd.com)	BD	AI text-based crop disease advisor (Bengali/English); bilingual blog on agriculture and AI	No image-based detection; no live prices; no user accounts
AgroSmart (agrosmartbd.com)	BD	AI crop recommendation based on soil, weather, and satellite data; pricing plans for farmers	No crop disease detection; no Bengali chatbot; no free tools
FlorianBD (florianbd.com)		CNN-based plant disease AI (95%+ accuracy); logistics optimization; smart recommendations	Focused on ornamental plants, not food crops; no market data integration

CropIQ is an entirely original platform, built from scratch, that addresses the identified gaps across all three platforms. It delivers a superior, unified, and free-to-access experience designed specifically for Bangladeshi food crop farmers, with features, architecture, and user experience that go beyond any single existing solution.

4. Features / Scope of the Project

In Scope (Will Be Built)

Feature Module	Description
AI Crop Disease Detector	Image upload → AI diagnosis → remedy suggestion (Bengali/English)
Live Market Price Board	Daily price data for 30+ commodities across BD districts
Weather + Crop Advisory	Location-based 7-day weather + AI planting/harvest advice
AI Farming Chatbot	Bengali-language Q&A chatbot using knowledge base + LLM
User Registration & Dashboard	Account creation, saved results, history, alerts
Bilingual UI (Bengali + English)	Full language toggle across the application

Out of Scope (Not in SDP Version)

- E-commerce / online marketplace functionality
- Native Android or iOS mobile application
- Custom machine learning model training (will use existing APIs)
- Payment gateway integration

5. System Architecture & Technology Stack

Layer	Technology	Justification
Frontend Framework	Next.js 14 (React)	Industry-standard framework; supports SSR for SEO; built-in API routes; widely documented for learners
Styling	Tailwind CSS	Utility-first CSS; fast development; fully responsive; supports Bengali fonts (Noto Serif Bengali)
Backend / API	Next.js API Routes + Python FastAPI	Next.js handles CRUD and auth; FastAPI handles AI inference (image processing) as a microservice
Database	PostgreSQL via Supabase	Free tier; includes authentication, real-time subscriptions, file storage, and pgvector extension for AI search
Authentication	Supabase Auth	Supports email/password and phone OTP — important for rural users without Google accounts
AI — Text Generation	Google Gemini 1.5 Flash API	Free API tier (15 req/min); strong Bengali language support; suitable for chatbot and advisory generation
AI — Image Analysis	Gemini Vision (multimodal)	Analyzes uploaded crop photos for disease identification; no custom model training required
Weather Data	OpenWeatherMap API	Free tier (1,000 calls/day); covers all 64 Bangladesh districts; 7-day forecast available
Market Price Data	Python web scraper (Playwright)	Automated nightly collection from DAM Bangladesh (govt price portal); stored in PostgreSQL
Hosting	Vercel (frontend) + Railway (backend)	Both have free tiers; Git-based auto-deployment; no server management required for SDP scope

6. Development Model

This project follows an Agile-inspired iterative development methodology divided into four phases, each with a concrete deliverable. This ensures steady, verifiable progress throughout the SDP timeline.

Why Agile?

- Allows for iterative development with regular deliverables
- Enables quick adjustments based on testing feedback
- Ensures steady, verifiable progress throughout the SDP timeline
- Facilitates collaboration among team members with different skill sets
- Reduces risk by delivering working features incrementally

Development Phases

Phase	Weeks	Activities	Deliverable
Phase 1: Foundation	Week 1–2	Project setup, database schema design, user auth, homepage UI, Vercel deployment	Live URL with registration and login working
Phase 2: Core AI Tools	Week 3–4	AI Crop Disease Detector (image upload + Gemini Vision); AI Farming Chatbot (Gemini text + knowledge base)	Two working AI tools accessible to users
Phase 3: Data Features	Week 5–6	Market Price Tracker (scraper + display); Weather Advisory (OpenWeatherMap + Gemini advisory generation)	Live price board and weather advisory functional
Phase 4: Polish & Testing	Week 7–8	User dashboard, mobile optimization, Bengali UI review, user testing with 20+ users, bug fixing	Complete deployable web application with user feedback report

7. Feasibility Analysis & Possible Risks

Technical Feasibility

All proposed technologies offer free tiers adequate for SDP-level traffic. Next.js and FastAPI are covered in standard CSE curricula. The Gemini API eliminates the need for custom model training, and Supabase removes the overhead of managing a separate database server.

Economic Feasibility

Resource	Cost
Vercel Hosting (Frontend)	Free
Supabase (Database + Auth + Storage)	Free (500MB tier)
Railway (Python Backend)	Free (\$5/month credit included)
Google Gemini API	Free (15 requests/minute)
OpenWeatherMap API	Free (1,000 calls/day)
Domain Name (.com)	~৳1,200/year (optional)
Total Estimated Cost	~৳1,200 for full SDP duration

Operational Feasibility

The target users — farmers and rural entrepreneurs — already use mobile internet. Bangladesh's 60%+ rural mobile penetration ensures platform reach. The bilingual interface and mobile-first design require no technical literacy to use.

Schedule Feasibility

The 8-week plan is realistic for a 6-member CSE team. Each phase produces a working, demonstrable output. The most complex components use ready-made AI APIs, keeping delivery well within the SDP timeline.

Possible Risks

- **API Rate Limits:** Free tiers of Gemini and OpenWeatherMap may have usage limits during peak testing
- **Data Reliability:** Market price scraper may break if the source website changes its structure
- **Internet Dependency:** Users in areas with poor connectivity may experience slow loading times
- **Language Accuracy:** AI-generated Bengali translations may require manual review and correction
- **Timeline Pressure:** Coordination among 6 team members may cause delays if responsibilities are unclear

8. Requirements

Hardware Requirements

- Development machines (6 units): 8GB RAM, 256GB SSD, Core i5 or equivalent
- Smartphones for testing (various Android and iOS devices)
- Network connectivity for testing and deployment

Software Requirements

- VS Code or similar IDE
- Node.js (v18+) and npm
- Python 3.9+
- Git and GitHub
- PostgreSQL (via Supabase)
- Browser Developer Tools

Personnel Requirements

- 6 development team members with distributed responsibilities
- Bengali language expert for UI review and content validation (optional)
- Agriculture subject matter expert for knowledge base validation (optional)

9. Project Breakdown with Time Estimation

	Stages/ Tasks	Efforts, man-hours
Stage 1	Analysis and Design	
1.1	Requirements analysis, mockup creation	12
1.2	Work plan creation	4
Stage 2	Implementation	
2.1	Web design creation (Homepage, Tools, Dashboard)	40
2.2	Database schema design and Supabase setup	20
2.3	User authentication and profile management	25
2.4	AI Crop Disease Detector (Gemini Vision integration)	30

	Stages/ Tasks	Efforts, man-hours
2.5	AI Farming Chatbot (knowledge base + Gemini API)	30
2.6	Market Price Scraper (Playwright + data pipeline)	24
2.7	Weather Advisory System (OpenWeatherMap + AI generation)	20
2.8	Bilingual UI (Bengali + English toggle)	16
2.9	User Dashboard and history tracking	20
Stage 3	Testing and other QA tasks	
3.1	Testing and bug fixing	24
3.2	User testing with 20+ users and feedback collection	16
3.3	Bengali UI review and refinement	8
Stage 4	Deployment	
4.1	Deployment on Vercel and Railway	5
4.2	Documentation and final report preparation	10
	Total estimated efforts: 294 man-hours	

Stage 1: Analysis and Design · Stage 2: Implementation · Stage 3: Testing and QA · Stage 4: Deployment.

10. Budget

The budget tentatively covers items such as employee salary, bonuses, benefits, training fees, professional fees, marketing materials, travel, office expenses, communication, hardware/software costs, and taxes, where applicable to a student project of this scale.

Project Item	Amount	Unit Price	Total
Software license	N/A	N/A	₳0
Machine(s) (Team's own devices)	6	₳0	₳0

Project Item	Amount	Unit Price	Total
Server License	N/A	N/A	৳0
Database license	Supabase Free Tier	৳0	৳0
Development Consultant	N/A	N/A	৳0
Project Management	N/A	N/A	৳0
Training (Time + Materials)	N/A	N/A	৳0
Domain Name (.com)	Optional	৳1,200/year	৳1,200
Total Estimated Budget			~৳1,200

11. Development Team & Stakeholders

Team Structure

ID	Name	Designation	Responsibilities
22235103255	Md. Torikul Islam Lipon	Project Lead / Full-Stack Developer	Overall architecture, Next.js frontend, API integration, deployment, project coordination
22235103127	Al-Amin Oyon	Backend Developer / AI Integration Specialist	Python FastAPI, Gemini API integration, image processing pipeline, chatbot backend
22235103236	Md Jobair Hossan	Database & DevOps Engineer	Supabase schema design, authentication setup, data seeding, deployment configuration, cron jobs
22235103028	Samia Homayara	UI/UX Designer & Frontend Developer	Bengali UI design, Tailwind CSS, mobile responsiveness, component library, design system
22235103230	Md. Aslam Hossain	Data Engineer / Web Scraper	Python Playwright scraper for market prices, data pipeline, knowledge base curation, RAG setup
22235103244	Sree Partho Chondro Mohanto	QA & Testing Lead	User testing coordination, mobile testing, Bengali UI review, bug reporting, documentation, user feedback collection

Stakeholders and Key Users

Stakeholder Type	Role	Interest in Project
Farmers	End users	Access free tools for disease diagnosis, market prices, and farming advice
Agriculture Students	End users	Learn about AI applications in agriculture; test and validate the system
Agricultural Extension Officers	Advisory users	Use the platform to assist farmers more efficiently
CSE Faculty (BUBT)	Supervisors	Evaluate project quality, provide guidance, and assess academic merit
General Public	Potential users	Access agricultural information and tools freely

12. Expected Outcomes

- A fully deployed, publicly accessible web application available at a live URL
- An AI Crop Disease Detector capable of identifying at least 10 common Bangladesh crop diseases from uploaded photos
- A daily-updated market price board covering at least 30 agricultural commodities across 10+ districts
- A weather-based crop advisory system operational for all 64 Bangladesh districts
- A Bengali-language AI chatbot capable of answering common farming questions accurately
- User account system with history tracking and preference saving
- A documented user testing report from at least 20 real users (farmers or agriculture students)

13. Limitations & Future Works

Limitations

- The AI disease detection accuracy depends on the quality of uploaded photos and may not detect all diseases with 100% confidence
- Market price data relies on third-party government sources and may have occasional delays
- Free tier API limits may restrict usage during peak hours
- The platform is web-based and requires internet connectivity (no offline mode)
- Language support is limited to Bengali and English only
- No custom machine learning model training; relies on existing APIs

Future Works

- Develop native Android and iOS mobile applications for better user experience
- Implement push notifications for weather alerts and market price changes
- Train a custom CNN model specifically for Bangladeshi crop diseases
- Add e-commerce/marketplace functionality for direct farmer-to-consumer sales
- Integrate with soil testing services and fertilizer recommendations
- Implement a community forum for peer-to-peer knowledge sharing
- Add support for additional regional languages (e.g., Chittagonian, Sylheti)
- Develop offline-first capabilities with cached data for low-connectivity areas

14. Conclusion

CropIQ addresses a real and measurable gap: Bangladeshi farmers lack a single platform for AI crop diagnosis, live market prices, weather advisory, and Bengali-language farming assistance. Built from scratch on a modern, freely deployable stack, this project delivers tangible social and economic value to an underserved population.

The project is technically, economically, and operationally feasible. Its modern stack (Next.js, Python, Gemini API, Supabase) can be built and maintained by the 6-member team with no specialized infrastructure and near-zero cost.

We believe this project represents a meaningful contribution to digital agriculture in Bangladesh and respectfully submit it for faculty approval as our Software Development Project.

15. References

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